

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2461183.8447 +/- 0.0033 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1361 +/- 0.01
Transit depth (Rp/Rs)^2	1.85 +/- 0.27 [%]
Orbital Inclination (inc)	89.48 +/- 2.15
Airmass coefficient 1 (a1)	1.515 +/- 0.019
Airmass coefficient 2 (a2)	-0.1941 +/- 0.0031
Scatter in the residuals of the lightcurve fit is	1.25 %
Best Comparison Star	None
Optimal Aperture	3.71
Optimal Annulus	10.01
Transit Duration (day)	0.0978 +/- 0.0017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1361 $\pm$ 0.01 vs 0.126 (deviation 1.01 σ)
Duration fit vs published	0.0978 d vs 0.0925 d (deviation 3.12 σ)
Mid-time O-C	5.2 min
Depth SNR	13.6
Residual scatter	1.25 %

Light curve

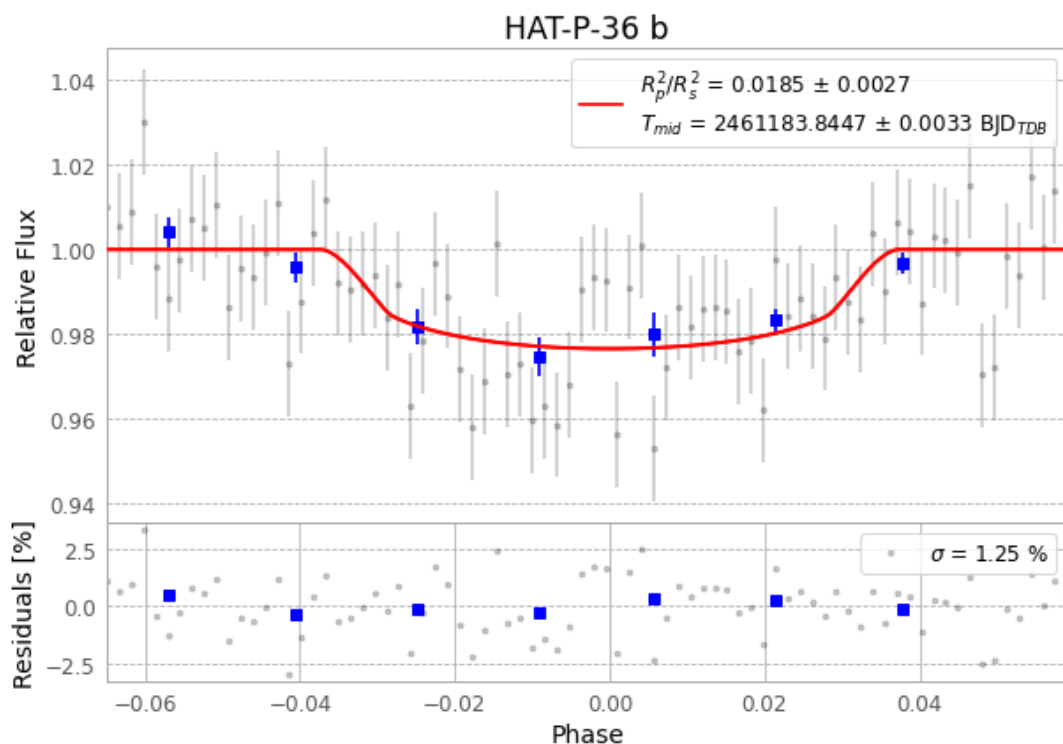


Figure 1 — FinalLightCurve_HAT-P-36 b_2026-05-22.png (SHA-256 65dc33268feb9448...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [264, 199], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9a651516038cbecc...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 62f7813

Data provenance

80 FITS frames (53 MB), HATP-36260523060916.FITS → HATP-36260523100615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-260523.log (040c0c519739...), FinalLightCurve_HAT-P-36 b_2026-05-22.png (65dc33268feb...), FinalLightCurve_HAT-P-36 b_2026-05-22.csv (986dcb8a6207...)

Compute resources

Wall time 165.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 18:52Z.